

JONATHAN KIM

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Education

The University of Texas at El Paso (UTEP)

Expected: December 2026

Bachelor of Science in Computer Science

GPA: 3.73

Relevant Coursework: Software Engineering, Databases, Object Oriented Programming, Data Structures & Algorithms

Experience

The University of Texas at El Paso

January 2025 – May 2025

Undergraduate Teaching Assistant - Elementary Data Structures & Algorithms

El Paso, Texas

- Mentored **30+ students** in **Java** to master data structures like linked lists and trees, raising lab completion rates and debugging accuracy across assignments.
- Resolved **20+ coding issues** weekly by teaching students to trace runtime errors and refactor redundant logic, **cutting repeat bugs by 30%**.

The Ohio State University AI-EDGE Institute

May 2024 – July 2024

Generative AI Researcher

Columbus, Ohio

- Optimized **Stable Diffusion 2** using **LoRA** training pipelines in **PyTorch**, processing 5 large datasets with **Pandas** to **reduce average training runtime by 18%** (10 h → 8 h 12 m).
- Fine-tuned **GPT-2** on a **1 M-token dataset** and evaluated outputs using **perplexity** and **BLEU** scores, achieving a **4% accuracy gain** and cleaner text coherence across test prompts.

Personal Projects

AeroScout | 2nd place — 49-team Hackathon

October 2025 – Present

- Designed an AI threat-assessment system using **InsightFace**, **YOLOv8**, and **React** to detect weapons and identify persons of interest from live drone footage with **100% accuracy**.
- Implemented backend risk-fusion logic in **FastAPI** and **OpenCV**, integrating facial and object detection results to deliver adaptive risk scores and event alerts within **under 200 ms per frame**.
- Developed an interactive **React** dashboard for live video streaming and situational alerts, improving response visualization and **decision support by 40%** during controlled test scenarios.
- Advancing development with partner companies, supported by **\$6K in funding** from JDC Energy Resource and a **\$5K custom drone** from PM Technologies, to transition AeroScout toward a deployable startup product.

Hada

May 2025 – Present

- Built a **React Native** fitness app for **iOS** and **Android** that records workouts and tracks streaks to encourage daily activity among users.
- Integrated **Firebase Authentication** and **Cloud Firestore** to securely sync data across devices, **maintaining 99% uptime** during testing.
- Validated app performance with **20+ beta users**, achieving a **65% daily check-in rate** and reliable data updates for streak analytics.

DUI Risk Radar | 2nd place — 22-team Hackathon

September 2024

- Led a team of **3** to develop a **React + Firebase** web app that maps DUI incident data and flags high-risk driving zones in real time.
- Implemented **Google Maps API** heatmaps to visualize accident-density data and simulate risk zones for testing.
- Constructed a reporting feature with **Google Cloud Speech-to-Text**, enabling **real-time transcription** and storage of incident reports with time and location in **Firebase**.

Leadership / Extracurricular

Free and Open-Source Club

January 2024 – January 2025

Secretary

UTEP

- Led **5+ hands-on workshops** on **Flask** and **Git**, **mentoring 15+ students** in deploying web apps and contributing to open-source repositories.

Technical Skills

Programming Languages: Python, Java, JavaScript, Swift, SQL

Frameworks & Tools: Git, HTML/CSS, React, SwiftUI, Flask, Node.js, Firebase

Machine Learning: PyTorch, Pandas